PHYC10009 Foundation of Physics – Lecture 1

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Physics, Room 501

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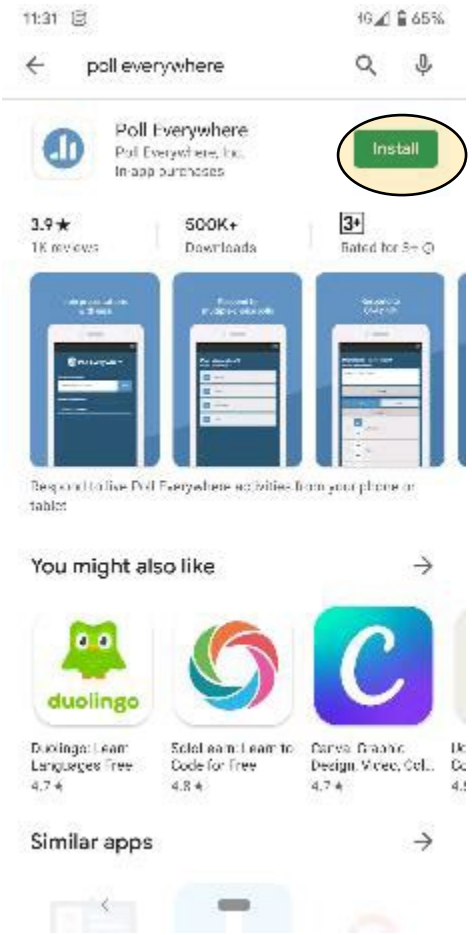
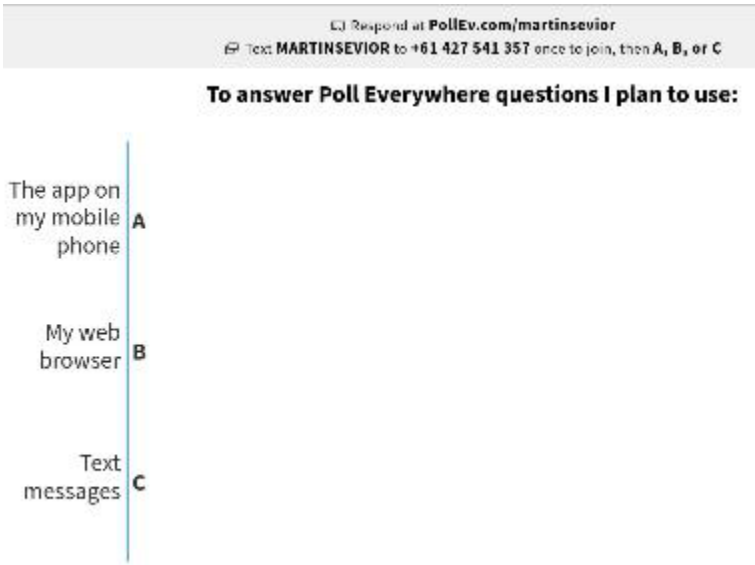


open

Web browser



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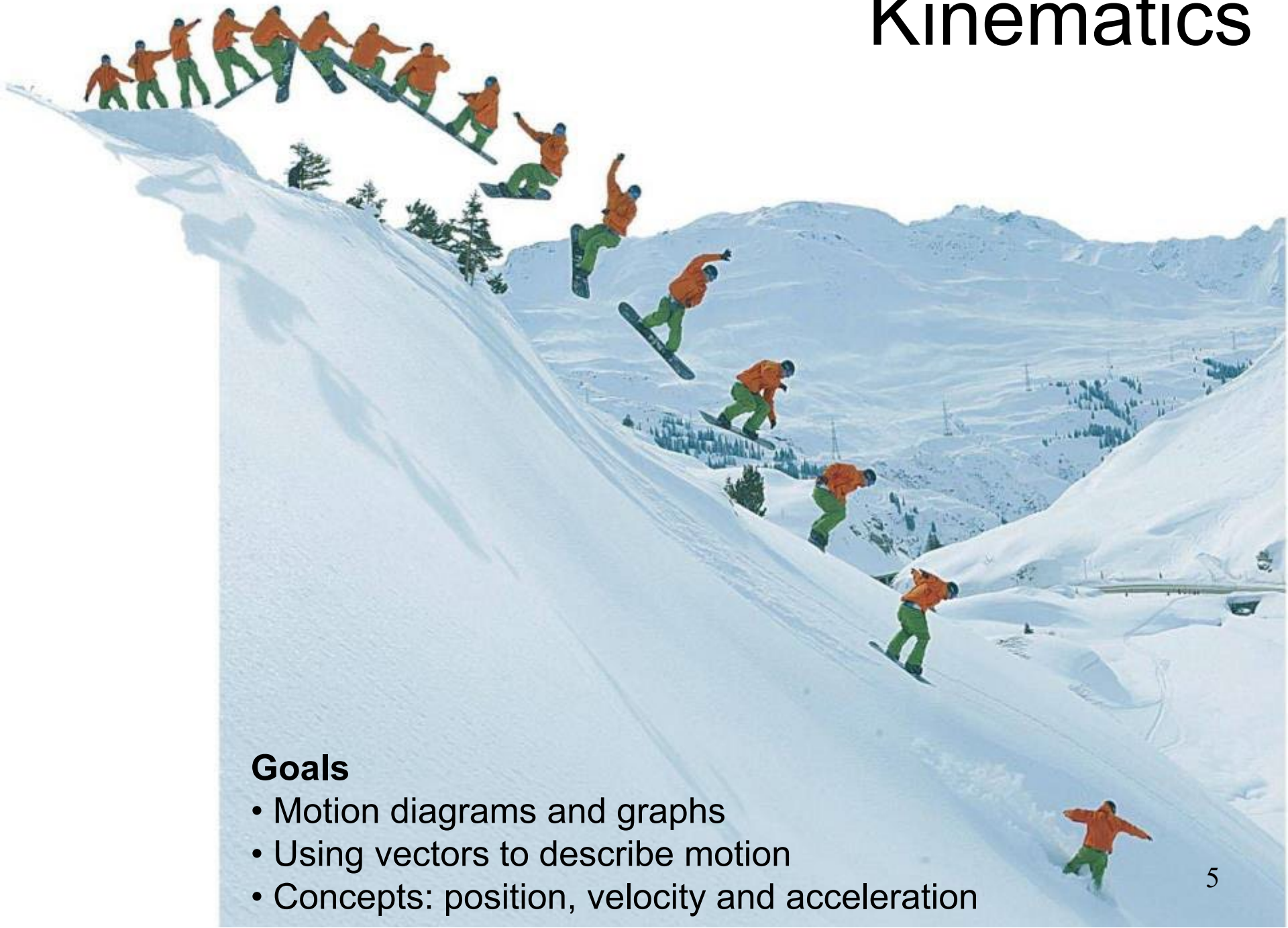
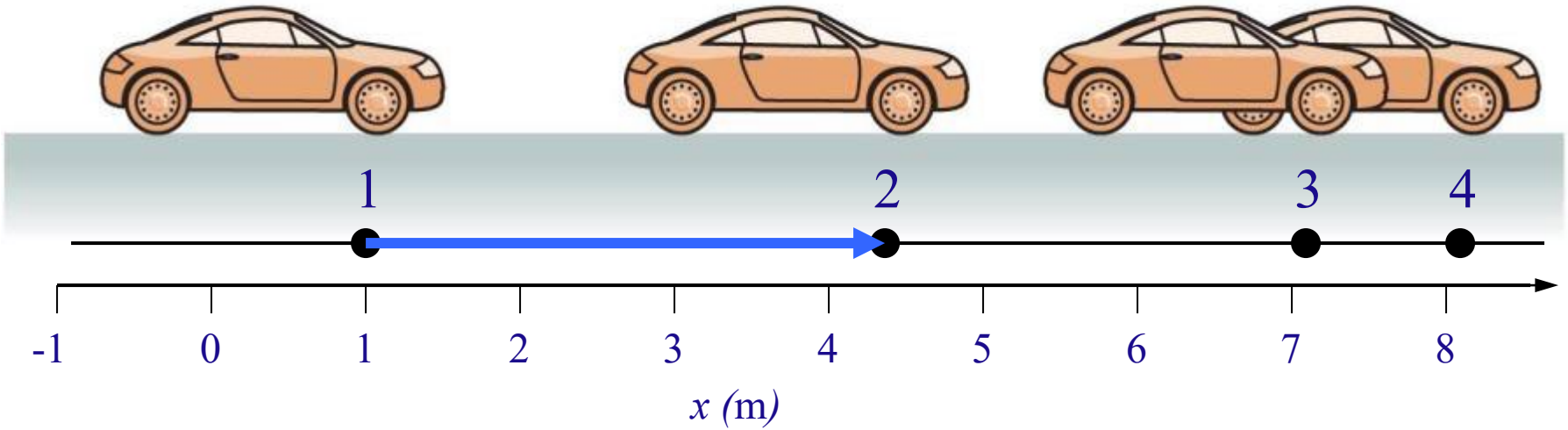
Kinematics



Goals

- Motion diagrams and graphs
- Using vectors to describe motion
- Concepts: position, velocity and acceleration

Displacement G2.1



Displacement is change in position:

$$\Delta x \equiv x_{final} - x_{initial}$$

e.g. Displacement of the car between frames 1 and 2 is

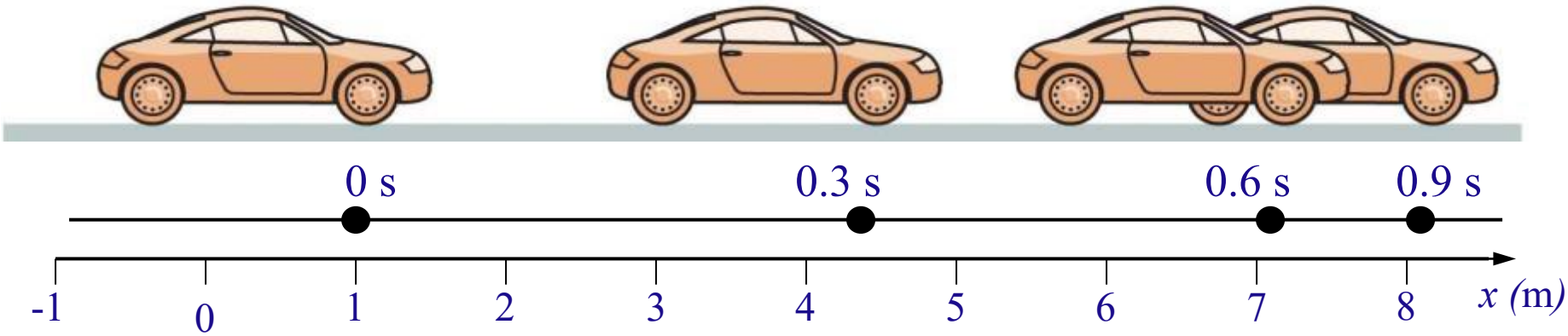
Δx = 4.3 m – 1.0 m = +3.3 m

(i.e. 3.3 m in the positive x-direction)

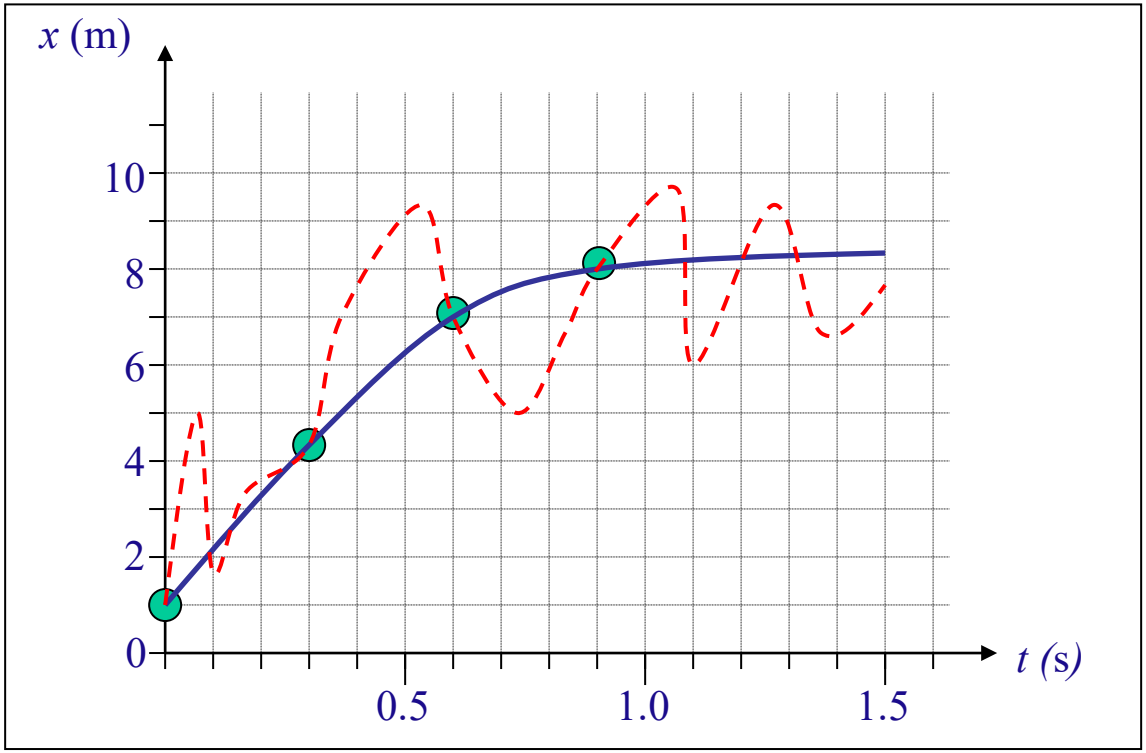
Note: Δx is a *single* symbol. Its value can be negative.

Position-time graph G2.2

Motion diagram



Position-time graph



1.3 Velocity - G 2.2

Speed: how fast

i.e. how much distance travelled in a given time interval

$$\text{Speed} = \frac{\text{distance travelled in a given time interval}}{\text{time interval}}$$

Displacement: change in position

$$\Delta x \equiv x_{final} - x_{initial}$$

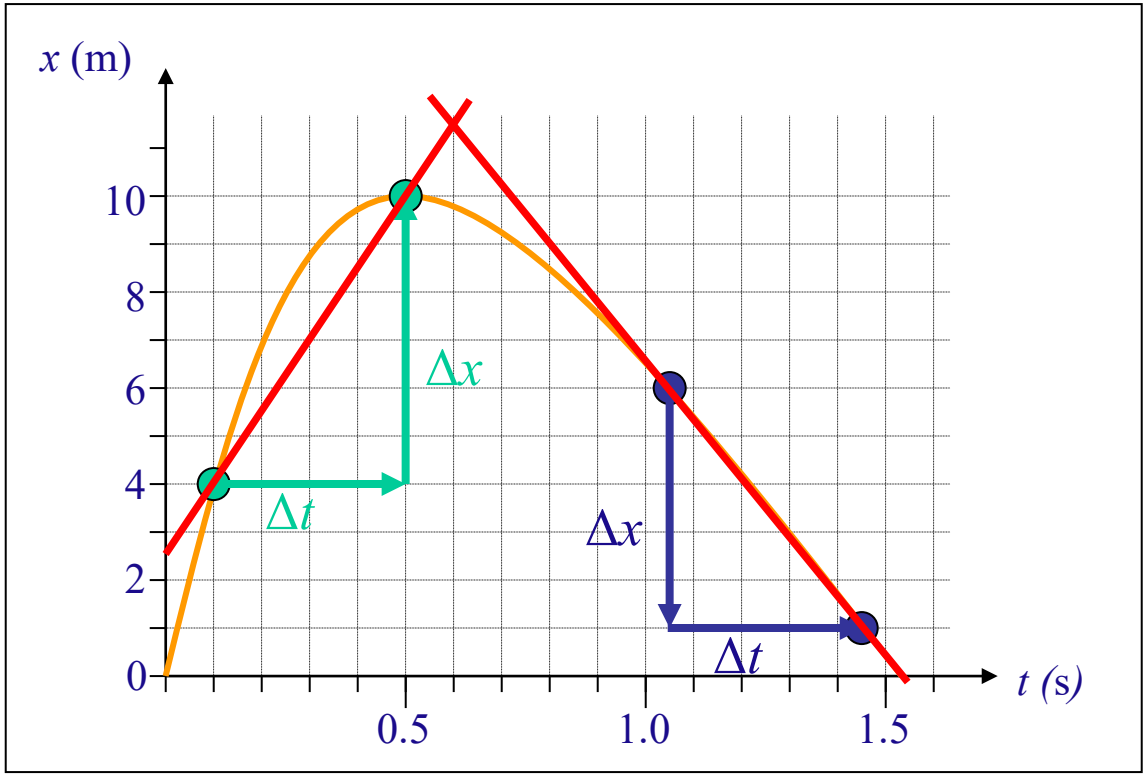
Velocity: how fast in a given direction

$$\text{Average velocity} = \frac{\text{displacement}}{\text{time interval}} = \frac{\Delta x}{\Delta t} = v_{av}$$

2.1 From position to velocity

$$v \equiv \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

Velocity is the *gradient* (*slope*) of position-time graph



Average velocity between t = 0.1 s and t = 0.5 s:

$$v_{av} = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i}{t_f - t_i}$$

$$= \frac{6\ \text{m}}{0.4\ \text{s}} = 15\ \text{m/s}$$

Average velocity between t =1.05s and t = 1.45 s:

$$v_{av} = \frac{\Delta x}{\Delta t}$$

$$= \frac{-5\ \text{m}}{0.4\ \text{s}} = -12.5\ \text{m/s}$$

Instantaneous velocity

Average velocity:

$$v_{av} \equiv \frac{\text{displacement}}{\text{time interval}} = \frac{\Delta x}{\Delta t}$$

To find velocity at an 'instant', choose a very small (infinitesimal) time interval

Instantaneous velocity:

$$v \equiv \lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t} = \frac{dx}{dt}$$

'Velocity' usually means 'instantaneous velocity', unless specified

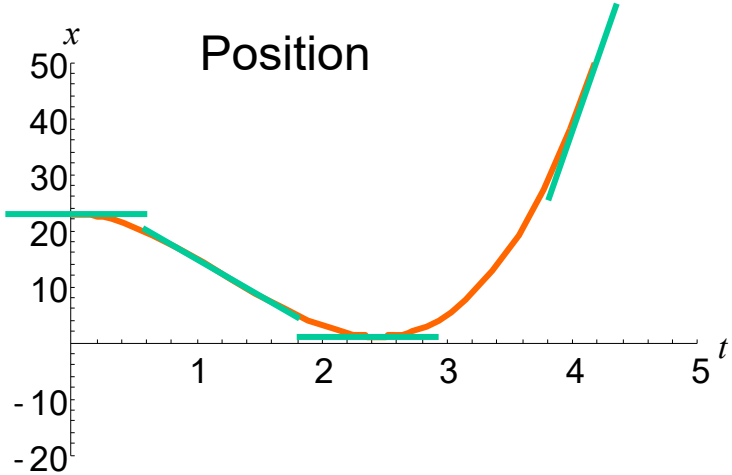
Differentiation and velocity G2.3

Velocity is *slope* (gradient) of position vs. time:

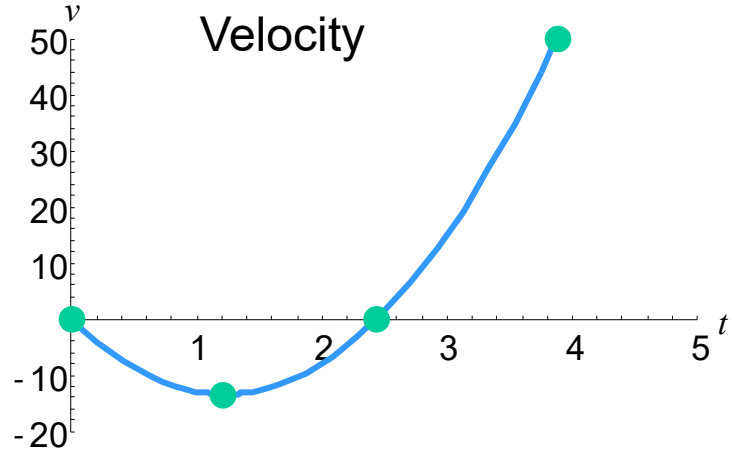
$$v \equiv \frac{dx}{dt}$$

Example: If we know $x(t) = 3t^3 - 11t^2 + 23$

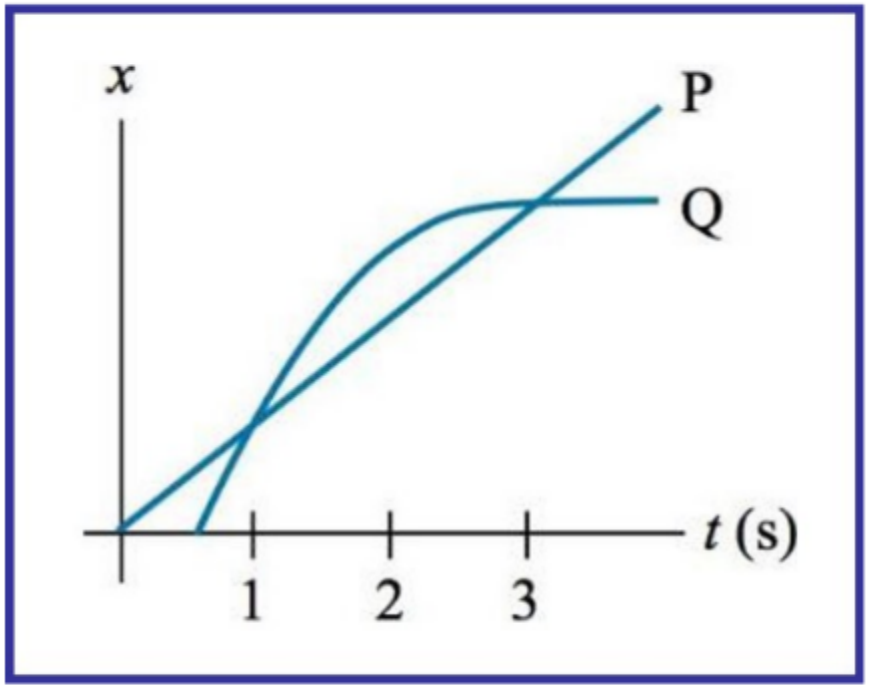
(x in metres, t in seconds)



Differentiate to find v: $v(t) = 9t^2 - 22t$



At which times(a) do particles P and Q have the same velocity?



At t=2s only

At t=1s and t=3s only

At t=1,2, and 3 s.

None of the above

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Changing velocity: acceleration - G 2.4

Recall velocity: rate of change of position $v \equiv \frac{dx}{dt}$

But velocity can also change

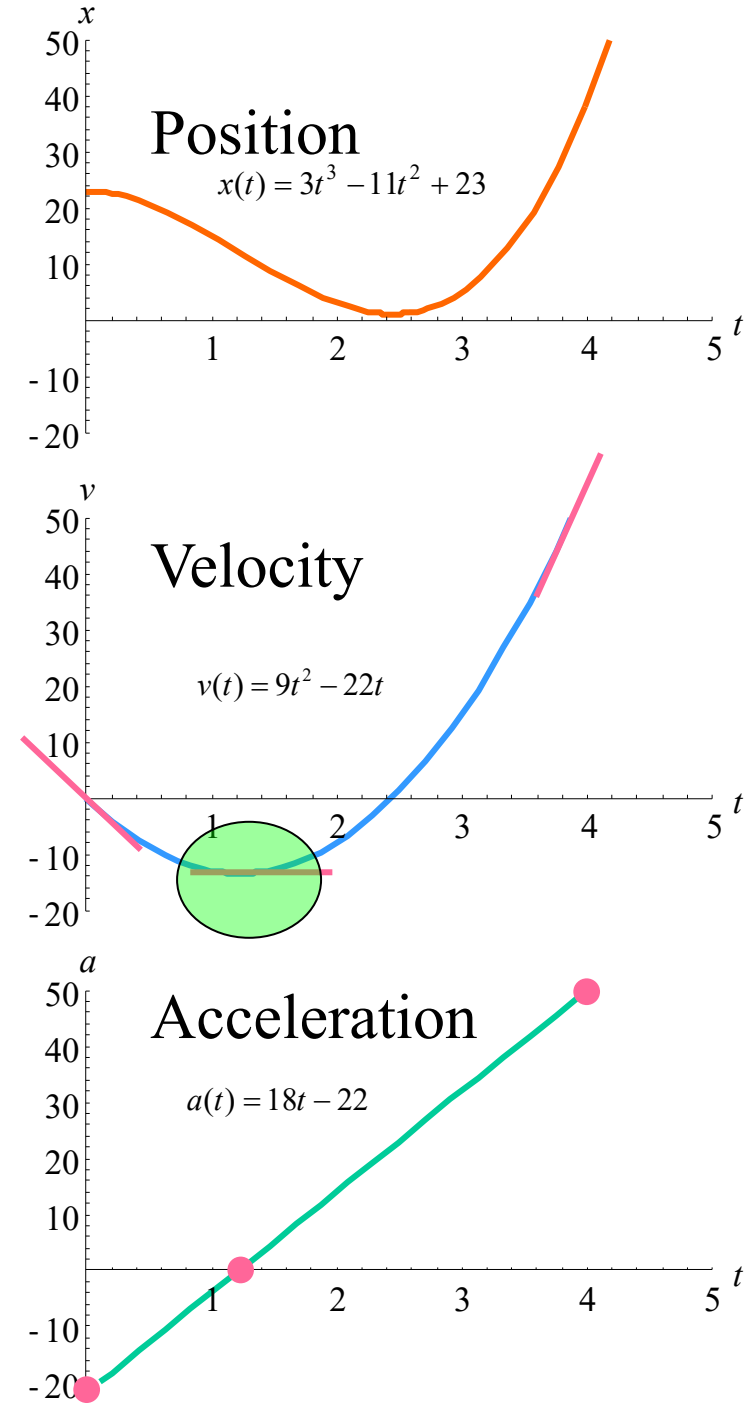
Consider earlier example (at right): velocity is changing – at nearly all times

*Question: when is the velocity **not** changing?*

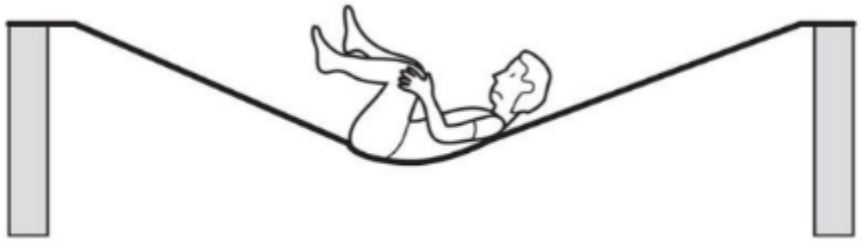
Define "how quickly velocity changes" (rate of change of velocity) as:

Acceleration $\boxed{a_{av} \equiv \frac{\Delta v}{\Delta t}}$ $\boxed{a \equiv \frac{dv}{dt}}$

$$v(t) = 9t^2 - 22t \quad \Rightarrow \quad a(t) = 18t - 22$$



Sam jumps out of a tree and lands on a trampoline.
The trampoline sags 60cm before launching Sam back into the air.
At the very bottom, where the sag is greatest, the direction of Sam's acceleration is:



Upwards

Downwards

Zero

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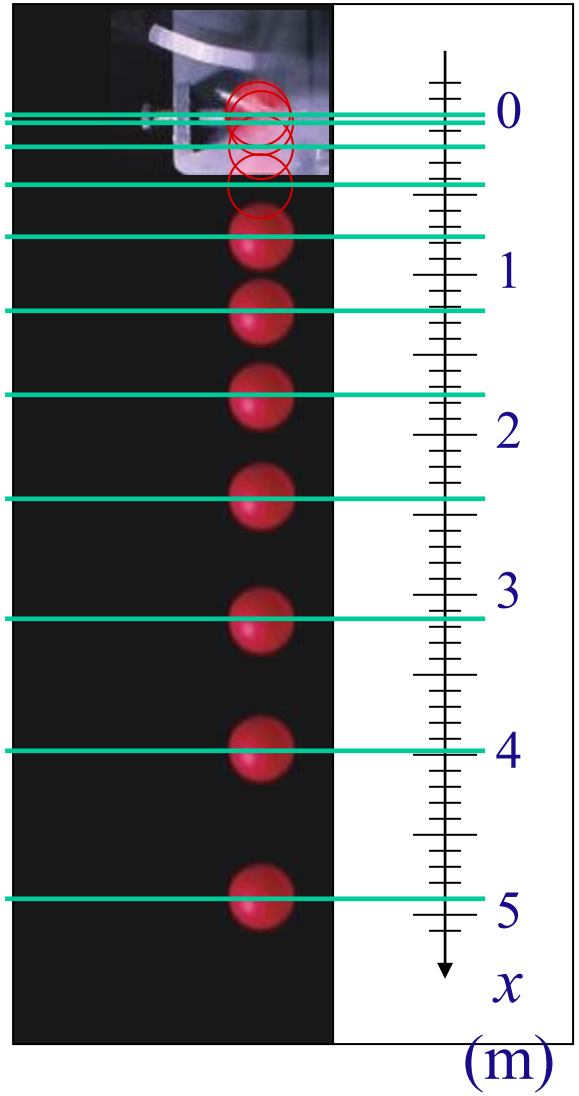
2.5 Constant acceleration G2.5

Free fall of a ball released from rest



t (s)	x (m)
0.0	0
0.1	0.049
0.2	0.196
0.3	0.441
0.4	0.784
0.5	1.225
0.6	1.764
0.7	2.401
0.8	3.136
0.9	3.969
1.0	4.900

Δx (m)	$v_{av} = \Delta x / \Delta t$ (m/s)	$a_{av} = \Delta v / \Delta t$ (m/s^2)
0.049	0.49	9.8
0.147	1.47	9.8
0.245	2.45	9.8
0.343	3.43	9.8
0.441	4.41	9.8
0.539	5.39	9.8
0.637	6.37	9.8
0.735	7.35	9.8
0.833	8.33	9.8
0.931	9.31	9.8

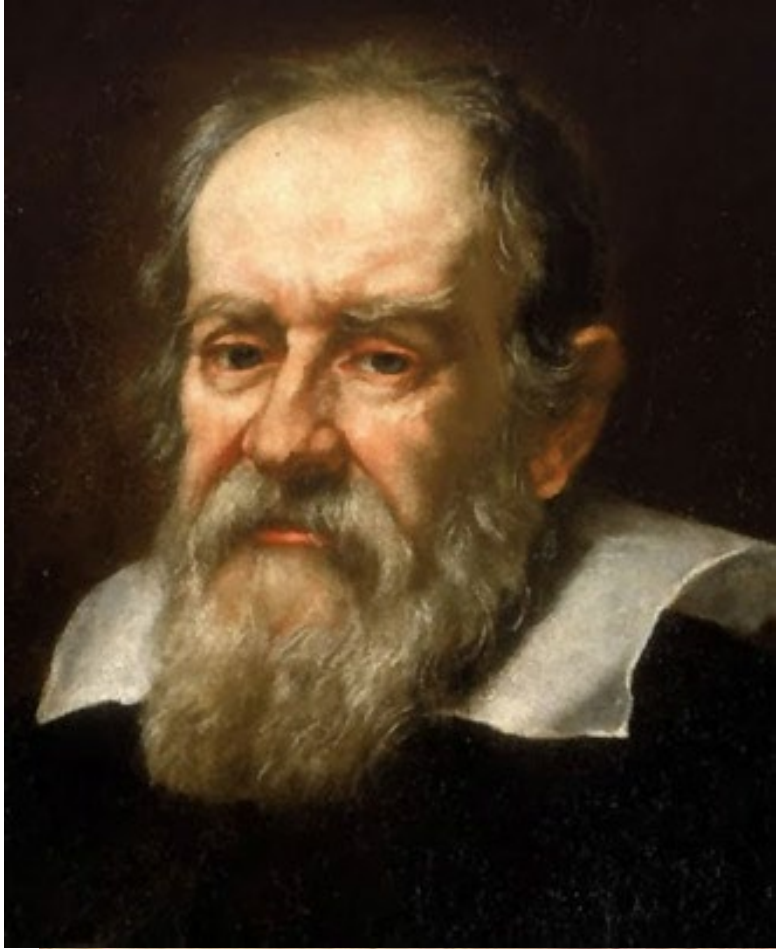
2.5 Constant acceleration G2.5

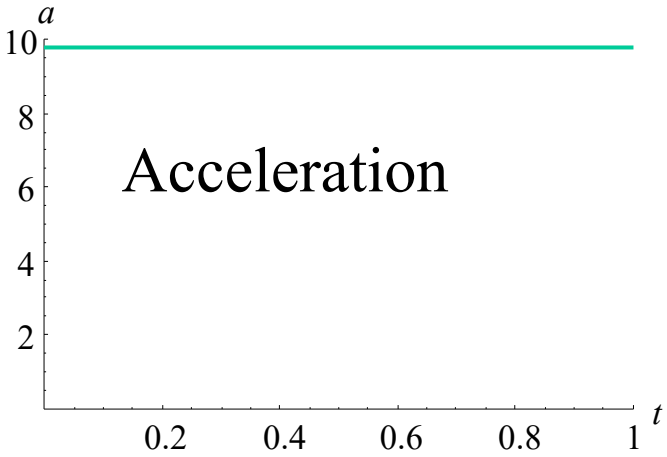
Gravity → constant downward acceleration of 9.80 m/s^2

We don't know the weight of the ball!

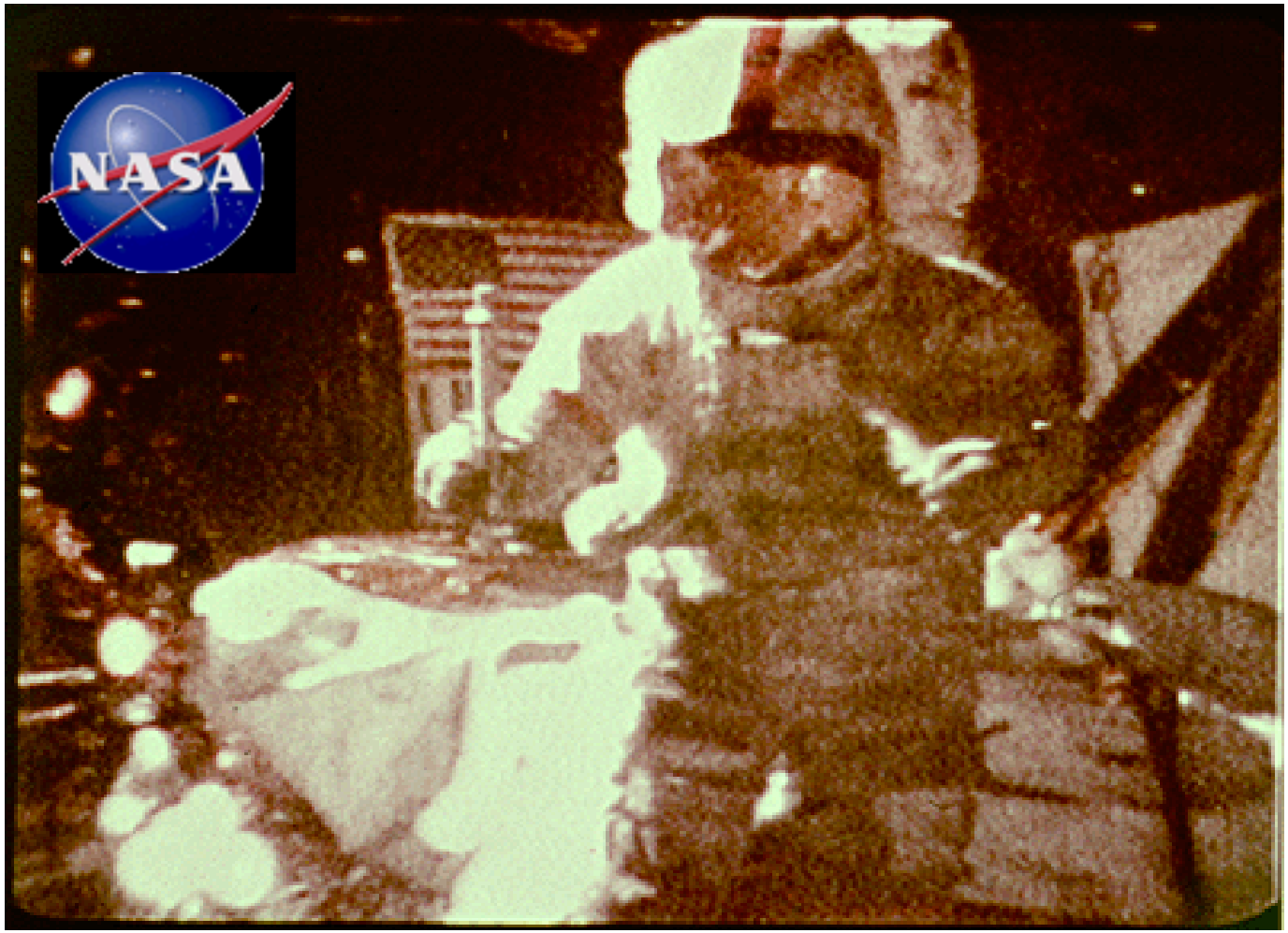
Aristotle: Heavier objects should fall 'faster' than light ones

Galileo: acceleration independent of mass





Free-fall demonstrations



Vacuum Feather demo

Apollo 15 Hammer and Feather Drop

http://nssdc.gsfc.nasa.gov/planetary/lunar/apollo_15_feather_drop.html
http://www.youtube.com/watch?v=5C5_dOEyAfk

Acceleration → Velocity → Position

If we know position…

$$x(t)$$

Velocity is *slope* of position vs. time:

$$v \equiv \frac{dx}{dt}$$

Acceleration is *slope* of velocity vs. time:

$$a \equiv \frac{dv}{dt}$$

Can we go back the other way?

$$a(t) \Rightarrow v(t) \Rightarrow x(t) \;\; ??$$

Consider walking (straight line) at 5km/hr

How far have you walked in ½ hr?

D'oh. 2.5km

You just *integrated the velocity-time function*

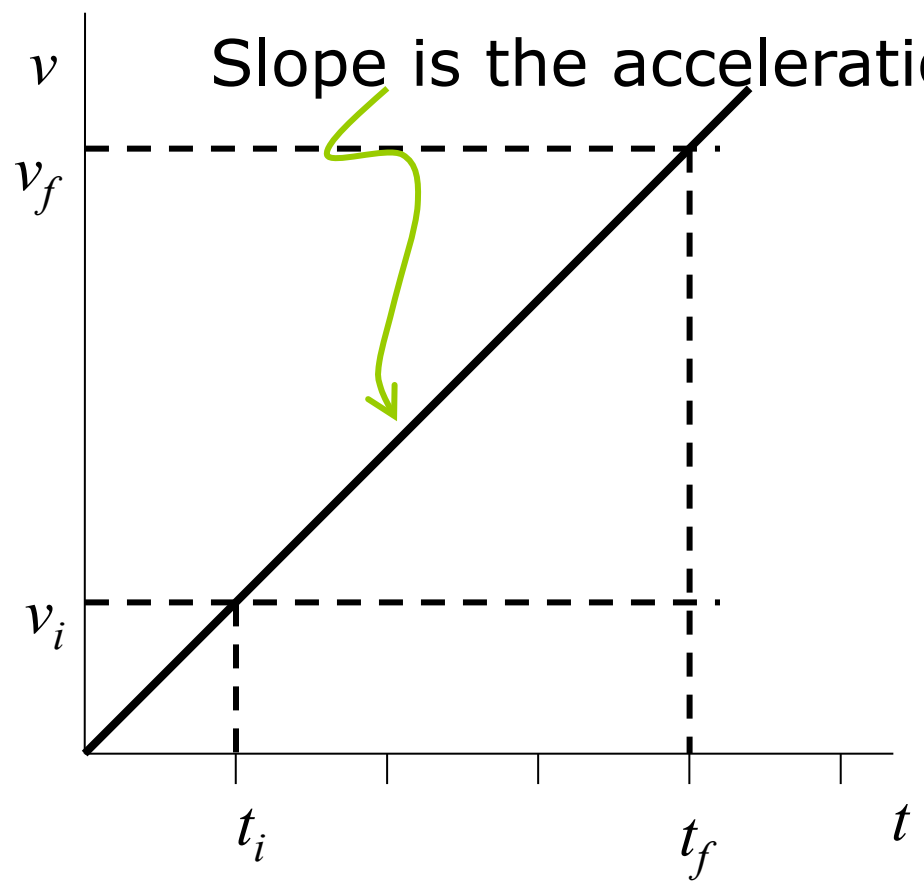
Get displacement by integrating velocity vs time function

Constant acceleration: velocity from acceleration

Very important special case:

Average acceleration is constant

$$a_{avg} \equiv \frac{\Delta v}{\Delta t} \quad \text{(average acceleration)}$$



That is,

$$a = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i}{\Delta t}$$

Rearrange: velocity is

$$\boxed{v_f = v_i + a\Delta t}$$

Constant acceleration: position from acceleration - G2.5

Velocity from acceleration

Integration is the **area** under a graph

$$v_f = v_i + a\Delta t$$

$$x_f = x_i + v\Delta t$$

Distance is velocity × time

$$\Delta x = v\,\Delta t$$

$$x = x_i + v\,\Delta t$$

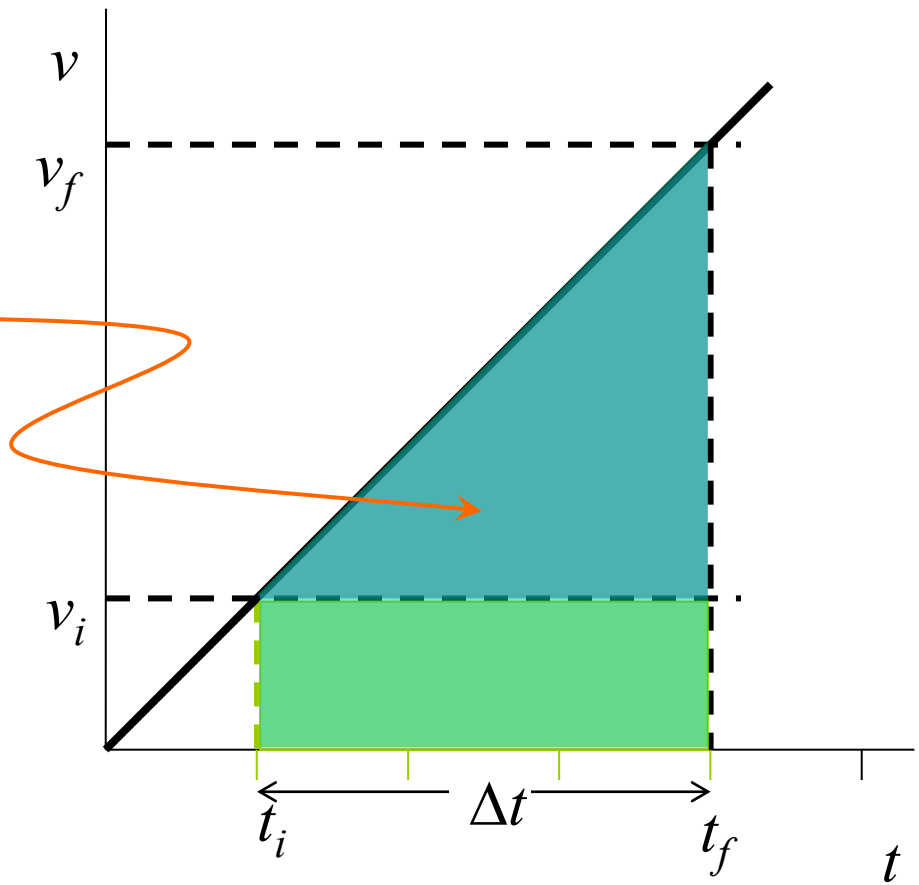
But if velocity is changing…

$$x = x_i + \sum v(t)\Delta t$$

That's just the area! i.e. integral…

$$x_f = x_i + \int v(t)\,dt$$

$$= x_i + v_i\Delta t + \tfrac{1}{2}a(\Delta t)^2$$



$x_f = x_i$ plus area under $v(t)$ curve

Equation of motion in 1d (constant acceleration)

For constant acceleration, we have $x_f = x_i + v_i \Delta t + \frac{1}{2} a (\Delta t)^2$

This is a critical formula!!

Initial position

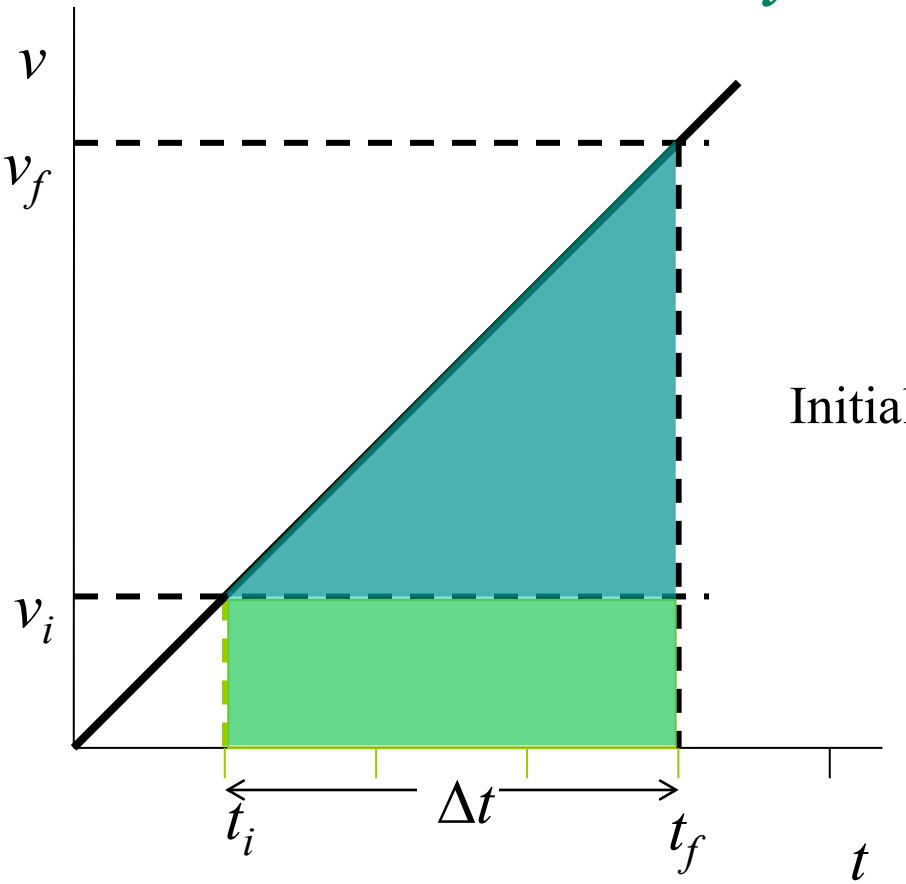
Initial velocity × time

Due to changing velocity

From this, can find velocity… $$v \equiv \frac{dx}{dt} = v_i + a\Delta t$$

and acceleration… $$\frac{dv}{dt} = a$$

Lecture 1 Summary



$$x_f = x_i + v_i \Delta t + \tfrac{1}{2} a (\Delta t)^2$$

Initial position

Initial velocity × time

Due to changing velocity

$$v \equiv \frac{dx}{dt} = v_i + a\Delta t$$

$$\frac{dv}{dt} = a$$